

Remittance Desk

System Documentation

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Live system: remittance-management-system-eight.vercel.app

1. Overview

Remittance Desk is a production web application that replaces spreadsheet-based tracking for a money-transfer business. It records money coming in from customers ("Income"), money sent out to beneficiaries on their behalf ("Sent Money"), and keeps a single, always-correct running balance — instead of a spreadsheet where the total is only as reliable as the last person who typed it in correctly.

The system is single-tenant: it runs one business's books. It supports two account roles — **Admin** and **Cashier** — so day-to-day transaction recording can be delegated to staff while account management and system configuration stay restricted to admins.

2. Technology Stack

- **Framework:** Next.js 16 (App Router), TypeScript, deployed on Vercel
- **Database:** PostgreSQL, hosted on Neon (serverless Postgres)
- **ORM:** Prisma, with raw SQL triggers for balance/audit logic (see §3)
- **Authentication:** Auth.js v5, email/password (Credentials provider), JWT sessions
- **UI:** Tailwind CSS, shadcn/ui components built on Base UI primitives
- **Charts:** Recharts
- **Forms/validation:** react-hook-form + Zod, shared between client forms and server actions
- **File storage:** Cloudflare R2 (S3-compatible) — receipt images, company logo
- **Report exports:** @react-pdf/renderer (PDF), ExcelJS (Excel), plain-text (CSV)

3. Core Architecture

Ledger-first balance

The current balance is never computed by summing every historical transaction on the fly. Instead, every completed transaction posts exactly one row to an append-only **ledger_entries** table via a database trigger — income credits the balance, sent transfers debit it. The current balance is simply the **balance_after** value on the single latest ledger row, which is an instant lookup rather than a recalculation. The transfer fee charged on a Sent Money transaction is tracked as reported revenue (visible in Reports) — it is not itself a ledger posting.

The ledger table is genuinely append-only: the database itself rejects any UPDATE or DELETE against it, at the trigger level — not just as an application convention that code could accidentally violate.

Exchange-rate snapshotting

Each Sent Money transaction stores the exchange rate that was in effect at the moment it was recorded. If the reference exchange rate is changed later in Settings, historical transactions keep showing the rate that actually applied to them.

Idempotency

Every income and sent-money submission carries a client-generated key. If a form is double-submitted (e.g. a double click, or a retry after a dropped connection), the database's uniqueness constraint on that key ensures the second attempt returns the original record instead of creating a duplicate transaction.

Audit logging

Changes to customers, income transactions, sent transactions, settings, and beneficiaries are logged automatically by a generic database trigger — not by application code remembering to call a logging function. This means a bug in a Server Action can never silently skip the audit trail. Login, logout, print, and export actions are logged directly by the application, since those aren't row mutations on an audited table.

4. Features

Dashboard

Current balance, today's income/sent/profit, a 7-day income-vs-sent chart, a monthly trend chart, top customers by income volume, and a feed of recent transactions.

Customers

Create, edit, search, and view a customer's full transaction history from their profile page.

Income

Record money received from a customer. Status workflow: pending -> completed (posts to the ledger) or cancelled. Sortable and searchable list, with a card view on mobile.

Sent Money

Record a transfer to a beneficiary on a customer's behalf. Includes:

- Beneficiary lookup by phone or name — matches an existing beneficiary automatically, or registers a new one on the spot
- Country/city cascading selection (Somalia and Djibouti today)
- Automatic 2.5% fee calculation from the amount, editable by hand if a transaction needs a different fee
- Optional receipt image attachment (JPEG/PNG/WEBP/HEIC, up to 8 MB)
- "Duplicate Transaction" — pre-fills a new Send Money form from a past one, for a repeat transfer to the same beneficiary

Reports

Filterable by period (today/week/month/year/custom), customer, and transaction type. Sortable by date, transaction number, or amount. Exports to CSV, Excel, and PDF, plus a printable browser view — all matching the on-screen summary (current balance, total income, total sent, total profit, transaction count).

Settings (admin only)

Manage reference exchange rates and user accounts (create/edit/deactivate/delete, assign Admin or Cashier role).

Mobile

A fixed bottom navigation bar and card-based lists replace the desktop sidebar and tables below tablet width, so the app is usable as a native-feeling mobile experience, not a shrunk desktop site.

Not yet built

- Receipts — a dedicated receipts module (PDF/QR/barcode generation) is planned but the page is currently a placeholder
- Audit Log viewer — the underlying audit trail is being recorded (see §3), but the page that displays it is currently a placeholder

5. Roles & Permissions

Capability	Admin	Cashier
Record income / sent transactions	Yes	Yes
View dashboard & customers	Yes	Yes
View & export reports	Yes	Yes
Create / edit customers	Yes	Yes
Delete a customer	Yes	No
Manage exchange rates	Yes	No
Manage user accounts	Yes	No
Access Settings	Yes	No

Role checks are enforced server-side on every restricted action (not just hidden in the UI), and a deactivated account is blocked on its very next request — deactivation does not wait for a token to expire.

6. Data Model

Key database tables:

- **users** — login accounts, role, active flag
- **customers** — the paying customer at the counter
- **beneficiaries** — recipients looked up/reused across Sent Money transactions
- **income_transactions** / **sent_transactions** — the two transaction types
- **ledger_entries** — append-only running balance (see §3)
- **audit_logs** — who changed what, and when
- **currencies** / **exchange_rates** — reference data for the Sent Money form
- **settings** — company name, default currency, and related configuration

7. Security

A full security audit was completed and its fixes deployed. Current posture:

In place

- Passwords hashed with bcrypt (cost factor 12)
- Every session is re-validated against the database on each request — a deactivated account loses access immediately, not just at next login
- Server-side role checks on every admin-only action, independent of what the UI shows
- All database queries are parameterized (no SQL injection surface)
- Every form is validated with Zod on the server, not just the client
- Security headers on every response: Content-Security-Policy, HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy
- File uploads are restricted to image types and capped at 8 MB, enforced by the storage provider itself
- No secrets are committed to source control; production credentials live only in Vercel's environment variables

Open items

- No rate limiting yet on login or API endpoints — planned, requires an Upstash Redis account to implement properly on serverless hosting
- No self-service "forgot password" flow — an admin resets a user's password via Settings
- Receipt images are reachable by anyone with the exact (unguessable) file URL, not behind a login check

8. Deployment

- **Hosting:** Vercel, auto-deploying from the GitHub repository's main branch on every push
- **Database:** Neon Postgres (production branch); schema migrations run automatically as part of every build, before the app compiles
- **File storage:** Cloudflare R2 — optional; the app runs fine without it, receipt/logo uploads simply stay unavailable until it's configured
- **Required environment variables:** DATABASE_URL, DIRECT_URL, AUTH_SECRET, NEXTAUTH_URL (plus the R2_* variables once file storage is set up)

9. Administration Guide

Add a staff account

Settings -> Users -> Add User. Choose Admin for full access, or Cashier for day-to-day transaction work only.

Change your own password

Settings -> Users -> edit your own row -> set a new password.

Add or update a reference exchange rate

Settings -> Exchange Rates. This only affects the Sent Money form's default suggestion — it never changes a rate already recorded on a past transaction.

10. Roadmap

- Rate limiting (Upstash Redis) on login and sensitive endpoints
- Receipts module — generated PDF receipts with QR/barcode
- Audit Log viewer page
- Cloudflare R2 configuration in production for receipt/logo uploads